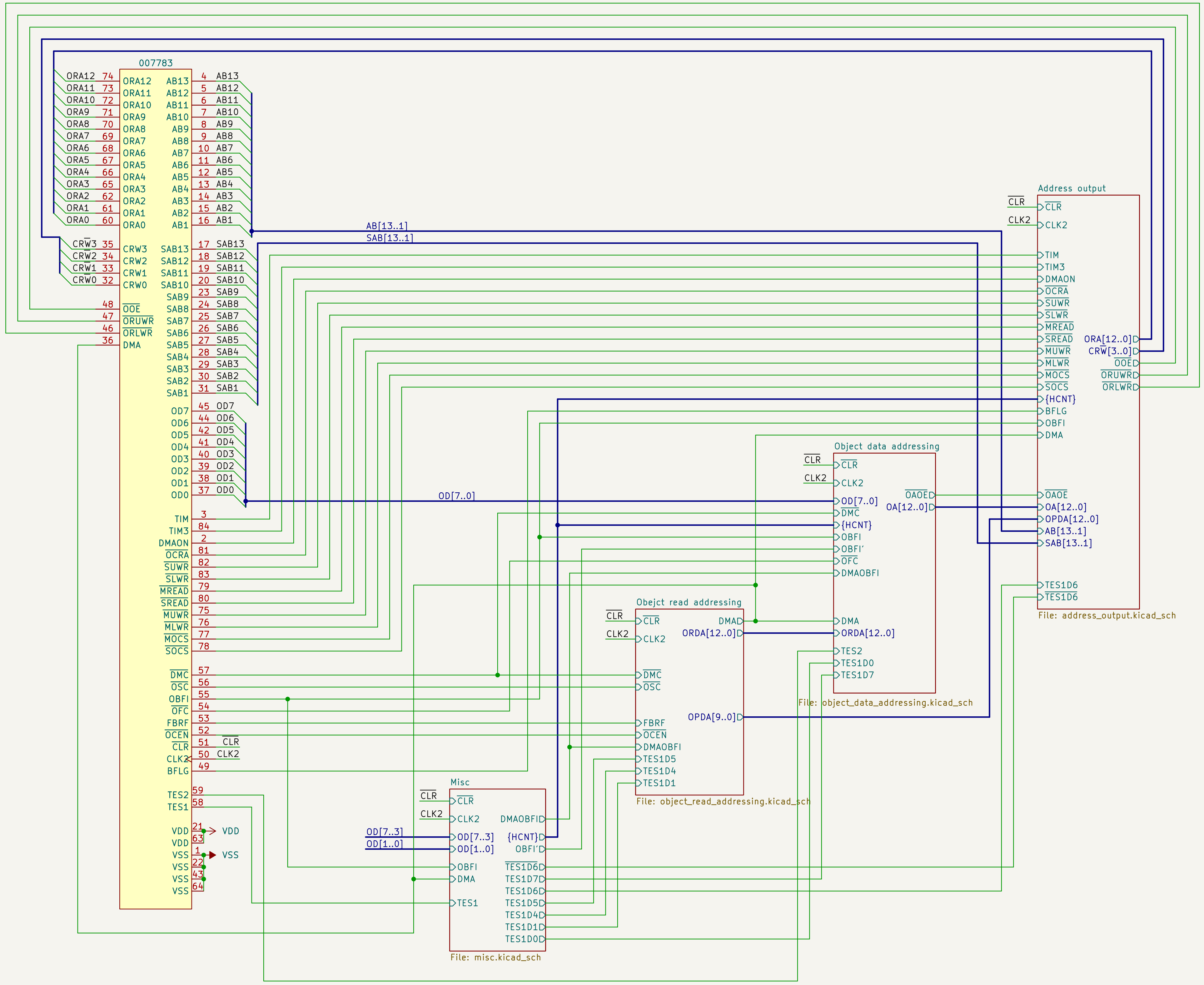
1. OBFI: (OBJBUFINIT) zeros are written to 1800h..1AFFh.
2. DMA: OA[12:0} active.
   a) 8 words are read from addresses 0000h..0007h, 0028h..002Fh,...,17E0h..17E7h
      by the sprite data parser 007784 and if the sprite is enabled with BEFLAG
      they are written to the buffer OBJBUF.

OBJRAM
0000h..17E7h Unprocessed data
1800h..1BFFh Processed data

WD0	WD1	WD2	WD3	WD4	WD5	WD6	WD7	WD0	WD1	WD2	WD3
0000	0001	0002	0003	0004	0005	0006	0007	BA1	BA2	BA3	BA4

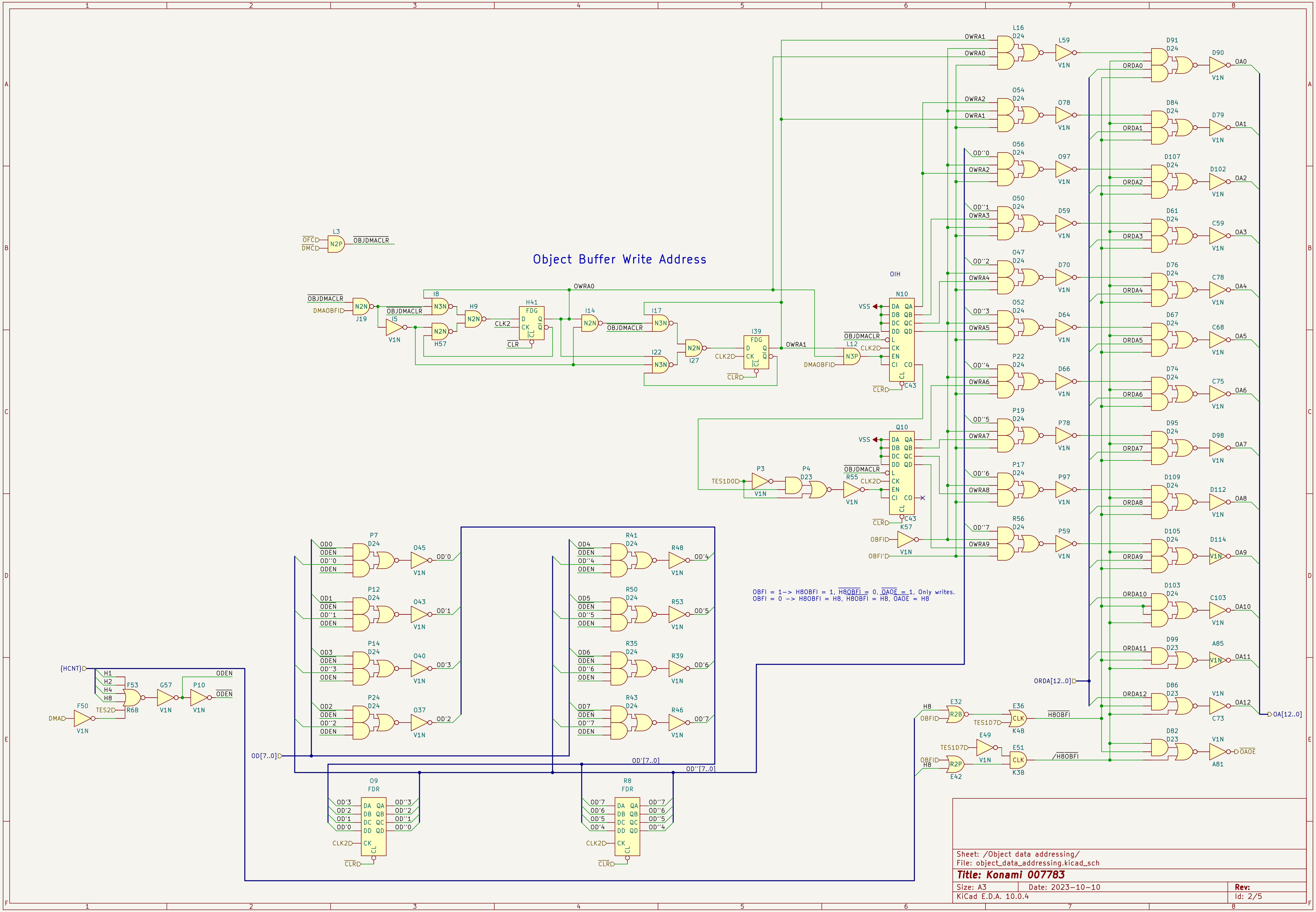
OBJBUF, Buffer Address
BA1 : OA[12:0] = 1800h + OD''[7:0] << 2
BA2 : OA[12:0] = BA1 + 1
BA3 : OA[12:0] = BA1 + 2
BA4 : OA[12:0] = BA1 + 3

OBFI	H8	OA	$\overline{\text{OAOE}}$	
1	X	{3'b110,OWRA[9:0]}	1	Writes to 1800h..1BFFh, reset object buffer
0	0	ORDA[12:0]	0	Read data from object RAM.
0	1	{3'b110,OD''[7:0],OWRA[2:1]}	1	Writes to 1800h..1BFFh



007783
ORA12 74, ORA11 73, ORA10 72, ORA9 71, ORA8 70, ORA7 69, ORA6 68, ORA5 67, ORA4 66, ORA3 65, ORA2 62, ORA1 61, ORA0 60
CRW3 35, CRW2 34, CRW1 33, CRW0 32
OOE 48, ORUWR 47, ORLWR 46, DMA 36
AB13 4, AB12 5, AB11 6, AB10 7, AB9 8, AB8 9, AB7 10, AB6 11, AB5 12, AB4 13, AB3 14, AB2 15, AB1 16
SAB13 17, SAB12 18, SAB11 19, SAB10 20, SAB9 23, SAB8 24, SAB7 25, SAB6 26, SAB5 27, SAB4 28, SAB3 29, SAB2 30, SAB1 31
OD7 45, OD6 44, OD5 42, OD4 41, OD3 40, OD2 39, OD1 38, OD0 37
TIM 3, TIM3 84, DMAON 2, OCRA 81, SUWR 82, SLWR 83, MREAD 79, SREAD 80, MUWR 75, MLWR 76, MOCS 77, SOCS 78
DMC 57, OSC 56, OBFI 55, OFC 54, FBRF 53, OCEN 52, CLR 51, CLK2 50, BFLG 49
TES2 59, TES1 58
VDD 21, VDD 63, VSS 1, VSS 22, VSS 43, VSS 64
VDD
VSS
CLR
CLK2
AB[13..1]
SAB[13..1]
OD[7..0]
Address output
CLR
CLK2
TIM, TIM3, DMAON, OCRA, SUWR, SLWR, MREAD, SREAD, MUWR, MLWR, MOCS, SOCS, {HCNT}, BFLG, OBFI, DMA
OAOE, OA[12..0], OPDA[12..0], AB[13..1], SAB[13..1]
TES1D6, TES1D6
ORA[12..0], CRW[3..0], OOE, ORUWR, ORLWR
File: address_output.kicad_sch
Object data addressing
CLR
CLK2
OD[7..0], DMC, {HCNT}, OBFI, OBFI', OFC, DMAOBFI
DMA, ORDA[12..0], TES2, TES1D0, TES1D7
OAOE, OA[12..0]
File: object_data_addressing.kicad_sch
Obejct read addressing
CLR
CLK2
DMA, ORDA[12..0]
DMC, OSC, FBRF, OCEN, DMAOBFI, TES1D5, TES1D4, TES1D1
OPDA[9..0]
File: object_read_addressing.kicad_sch
Misc
CLR
CLK2
OD[7..3]
OD[1..0]
OD[7..3], OD[1..0], OBFI, DMA, TES1
DMAOBFI, {HCNT}, OBFI', TES1D6, TES1D7, TES1D6, TES1D5, TES1D4, TES1D1, TES1D0
File: misc.kicad_sch

Ulf Skutnabba, twitter: @skutis77
Sheet: /
File: 007783.kicad_sch
Title: Konami 007783
Size: A3 Date: 2023-10-10 **Rev:**
KiCad E.D.A. 10.0.4 Id: 1/5


Object Buffer Write Address
L3
N2P
OFC
DMC
OBJDMACLR
DMAOBFI
J19
N2N
I8
N3N
H57
I5
V1N
H9
H41
FDG
CLK2
CLR
OWRA0
I14
I17
I22
I27
I39
OWRA1
L12
N3P
OIH
N10
VSS
DA DB DC DD
QA QB QC QD
L CK EN CI CO CL
C43
Q10
TES1D0
P3
P4
D23
R55
K57
OBFI
OBFI'
L16
D24
L59
O54
O78
O56
O97
O50
D59
O47
D70
O52
D64
P22
D66
P19
P78
P17
P97
R56
P59
OWRA2 OWRA3 OWRA4 OWRA5 OWRA6 OWRA7 OWRA8 OWRA9
OD''0 OD''1 OD''2 OD''3 OD''4 OD''5 OD''6 OD''7
D91 D90 D84 D79 D107 D102 D61 C59 D76 C78 D67 C68 D74 C75 D95 D98 D109 D112 D105 D114 D103 C103 D99 A85 D86 C73 D82 A81
ORDA0 ORDA1 ORDA2 ORDA3 ORDA4 ORDA5 ORDA6 ORDA7 ORDA8 ORDA9 ORDA10 ORDA11 ORDA12
OA0 OA1 OA2 OA3 OA4 OA5 OA6 OA7 OA8 OA9 OA10 OA11 OA12
ORDA[12..0]
OA[12..0]
H8
E32
R2B
E36
CLK
K4B
TES1D7
H8OBFI
E49
E51
K3B
R2P
E42
OAOE
P7 O45 P12 O43 P14 O40 P24 O37
R41 R48 R50 R53 R35 R39 R43 R46
OD0 OD1 OD2 OD3 OD4 OD5 OD6 OD7
ODEN
OD'0 OD'1 OD'2 OD'3 OD'4 OD'5 OD'6 OD'7
OD[7..0]
OD'[7..0]
OD''[7..0]
{HCNT}
H1 H2 H4 H8
F53
R6B
G57
P10
F50
DMA
TES2D
O9
R8
FDR
1 2 3 4 5 6 7 8
A B C D E F

OBFI = 1-> H8OBFI = 1, $\overline{\text{H8OBFI}}$ = 0, $\overline{\text{OAOE}}$ = 1, Only writes.
OBFI = 0 -> H8OBFI = H8, H8OBFI = H8, OAOE = H8

Sheet: /Object data addressing/
File: object_data_addressing.kicad_sch

Title: Konami 007783

Size: A3 | Date: 2023-10-10 | Rev:
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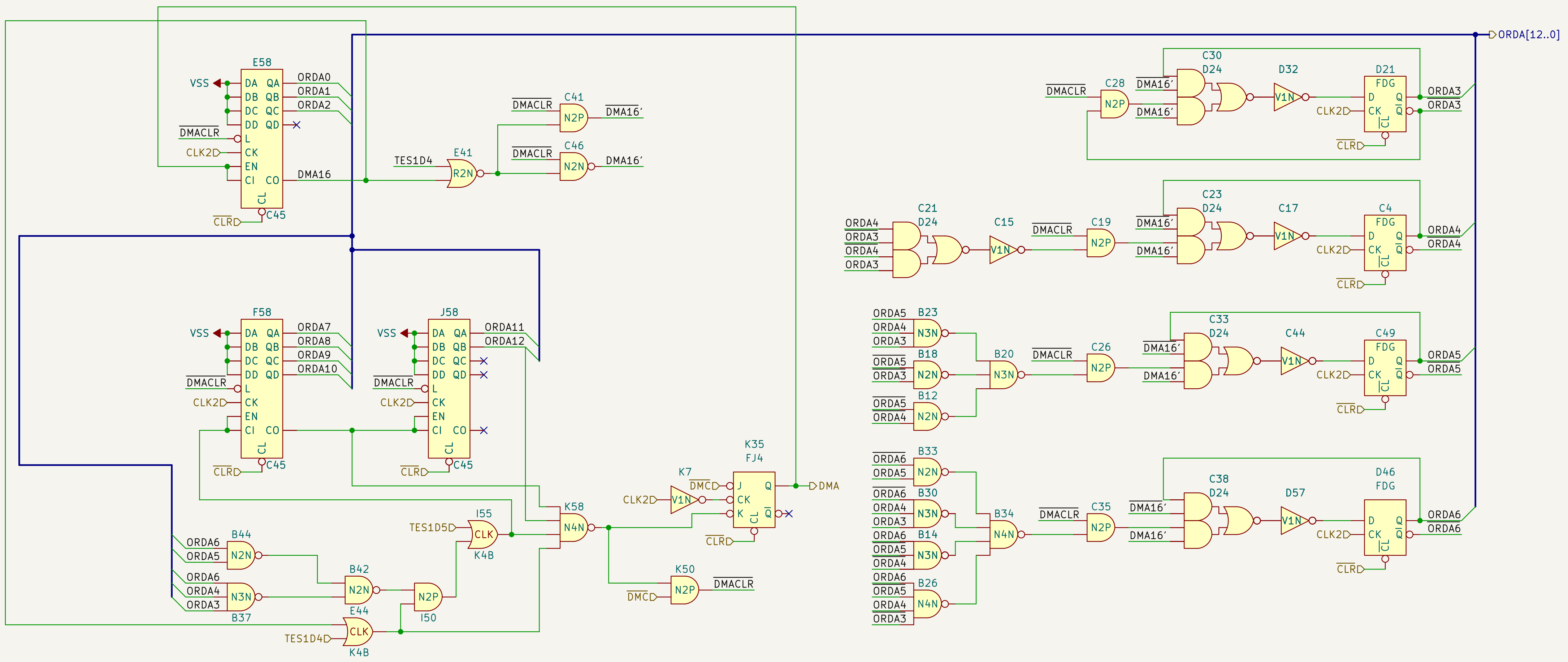
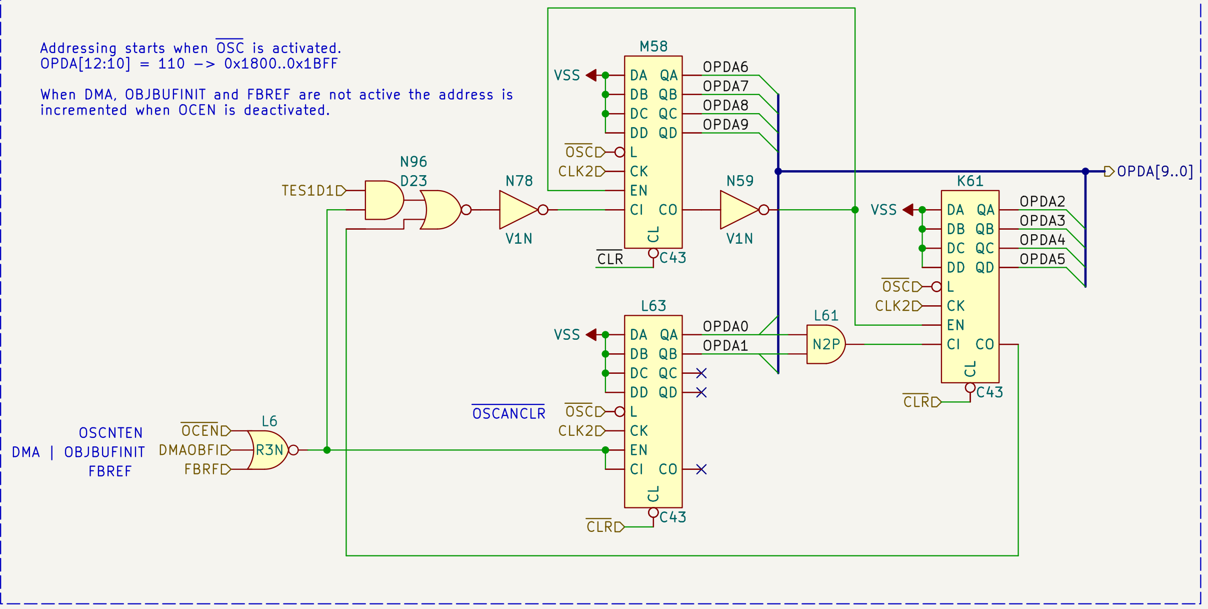
Object Read Addressing



Parsed Object Data Addressing



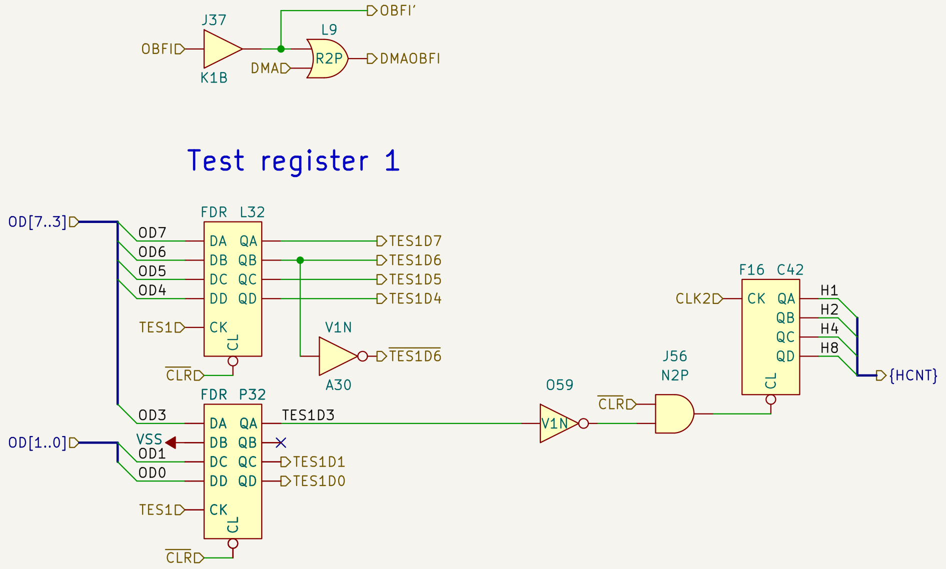
Addressing starts when $\overline{OSC}$ is activated.
OPDA[12:10] = 110 -> 0x1800..0x1BFF

When DMA, OBJBUFINIT and FBREF are not active the address is incremented when OCEN is deactivated.

Sheet: /Obejct read addressing/
File: object_read_addressing.kicad_sch

Title: Konami 007783

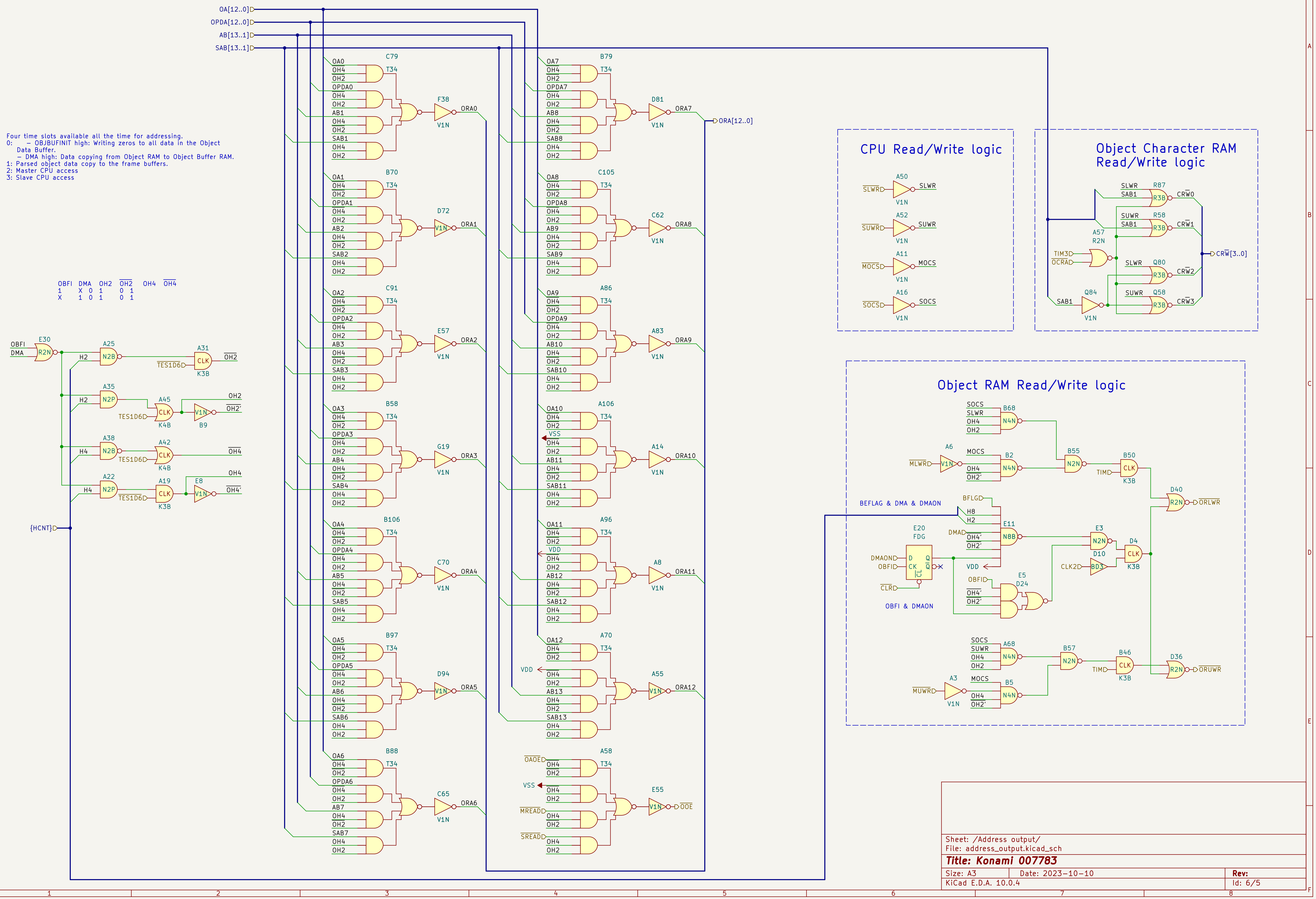
Size: A3 Date: 2023-10-10 Rev:
KiCad E.D.A. 10.0.4 Id: 3/5



Sheet: /Misc/
File: misc.kicad_sch

Title: Konami 007783

Size: A3 Date: 2023-10-10 **Rev:**
KiCad E.D.A. 10.0.4 Id: 5/5



Four time slots available all the time for addressing.
0: – OBJBUFINIT high: Writing zeros to all data in the Object Data Buffer.
– DMA high: Data copying from Object RAM to Object Buffer RAM.
1: Parsed object data copy to the frame buffers.
2: Master CPU access
3: Slave CPU access

OBFI	DMA	OH2	$\overline{OH2}$	OH4	$\overline{OH4}$
1	X	0	1	0	1
X	1	0	1	0	1

Sheet: /Address output/
File: address_output.kicad_sch

Title: Konami 007783

Size: A3 Date: 2023-10-10 **Rev:**
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